

Bridge Report

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1.Introduction (general background information about trusses and bridges)

Trusses are structures composed of triangular units that connect with each other at joints. They're commonly used to construct structurally sound pieces of architecture such as bridges as they're extremely cost efficient, strong and consume less resources as they span long distances. This makes them extremely useful in modern day bridges as vehicles such as cars with passengers or lorries delivering products exert heavy load onto the bridge yet are still able to get to their destinations more efficiently and sustainably than they otherwise wouldn't be if our bridges weren't structurally secure thanks to truss implementation.

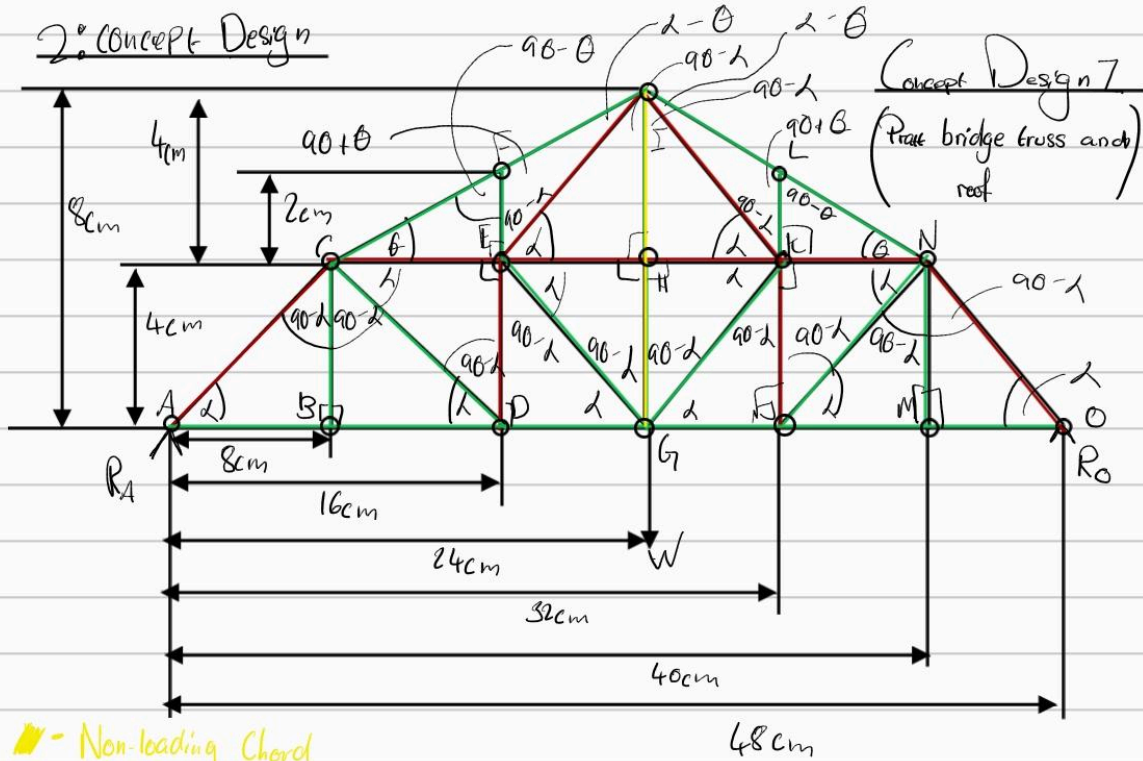
In other words, trusses and bridges play an important factor in the lives of most people as our demand for products increases and the growing population causing an increased demand for transportation services means that we're going to be growing more and more reliant on these structures to sustainably fulfill the needs of future generations.

Hence forth, the aim of our project is to see whether we can create such a model truss bridge where said bridge would need to cross a span of 400mm between rigid cliffs on either side of a river, allow the free passage of a single lorry 60mm wide and 70mm high, have an easy access of a 50x50mm helicopter landing pad at the centre of the bridge and lastly, withstand a load of 60N.

2. Concept design (description of various designs considered, including sketches, diagrams, etc.).

Figure 1 and 2 (concept truss bridges)

2nd Concept Design



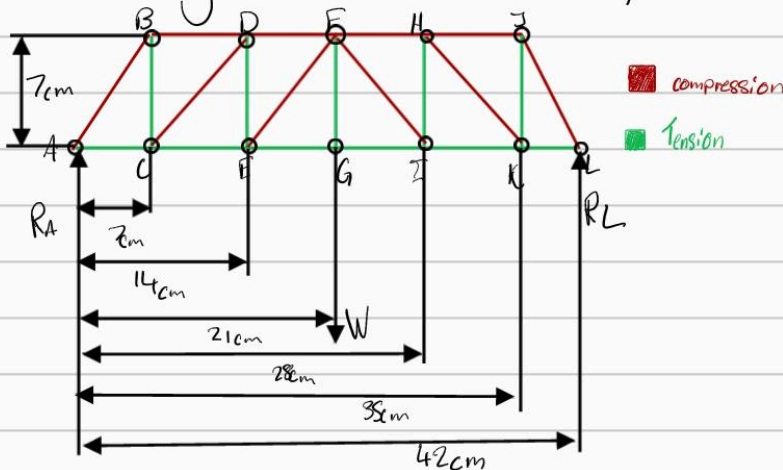
■ - Non-loading Chord

■ - Tension

■ - Compression

We decided to abandon this design as the roof might add too much weight to the bridge, which decreases the load to mass ratio of the bridge, making it less efficient. Furthermore, the Pratt truss roof on top of the Pratt truss bridge increases the cost of resources used to build it as discarding the roof and altering the vertical height of the bridge to be a minimum of 70mm would make things easier to construct and decrease the weight of the bridge.

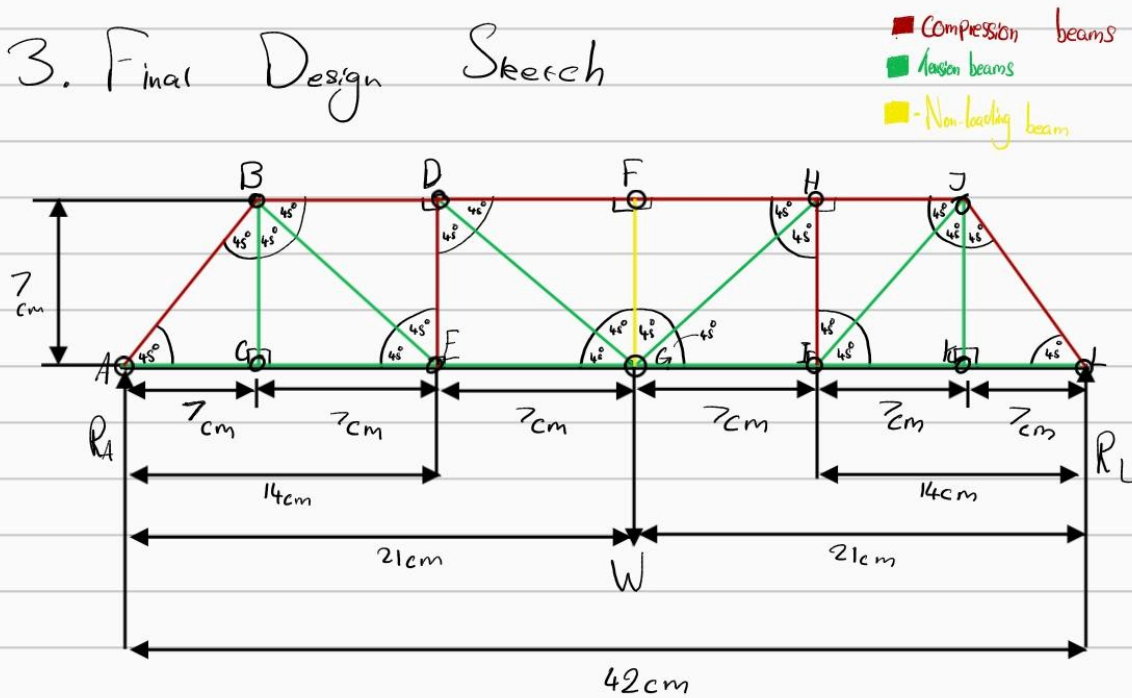
Concept Design 2 (Howe Bridge Truss)



This Howe truss bridge design was also discarded due to the fact that the longer diagonal ^{and upper horizontal} members are in compression whereas the shorter vertical members and lower horizontal members are in tension. This means the bridge is more likely to collapse as the overall compression would be greater than the overall tension.

Figure 3 and 4 (Final bridge design and calculations)

3. Final Design Sketch



4. Calculations

(Negative Sign = compression, Positive Sign = Tension)

Moments about point L:

Moments about A:

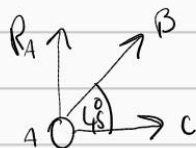
$$\frac{42 R_A}{42} = \frac{21 W}{42} = \frac{W}{2} N$$

$$\frac{42 R_L}{42} = \frac{21 W}{42} = \frac{W}{2} N$$

Horizontal resultant $\propto$ ON

Vertical resultant force: $R_A + R_L = W N$

Conditions of equilibrium at A



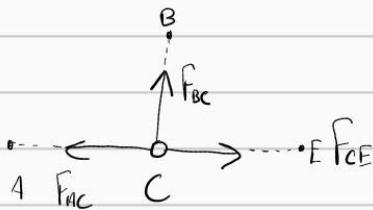
$$x: F_{AB} \cos(45^\circ) + F_{AC} = 0 N$$

$$y: R_A + F_{AB} \sin(45^\circ) = 0 N$$

$$F_{AC} = -F_{AB} \cos(45^\circ) = - \times \frac{W \sqrt{2}}{2} \times \cos(45^\circ) = \frac{W}{2} N$$

$$F_{AB} = \frac{-R_A}{\sin(45^\circ)} = \frac{-W}{2} \times \sqrt{2} = \frac{-W \sqrt{2}}{2} N$$

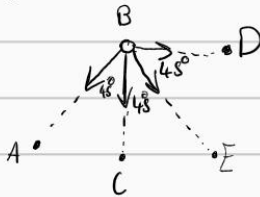
Conditions of equilibrium at C



$$x: F_{CE} - F_{AC} = \frac{W}{2} N$$

$$y: F_{CB} = 0 N$$

Conditions of equilibrium at point B



$$x: F_{BD} + F_{BE} \cos(45^\circ) = F_{BA} \sin(45^\circ) N$$

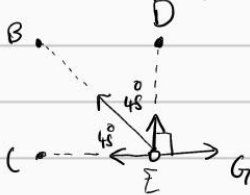
$$y: -F_{BE} \cos(45^\circ) - F_{BC} - F_{BA} \cos(45^\circ) = 0 N$$

$$x: F_{BD} + F_{BE} \cos(45^\circ) = -\frac{W}{2} N$$

$$y: F_{BE} = \frac{-F_{BC} - F_{BA} \cos(45^\circ)}{\cos(45^\circ)} = \frac{-0 - x - \frac{W\sqrt{2}}{2} \cos(45^\circ)}{\cos(45^\circ)} = \frac{W\sqrt{2}}{2} N$$

$$F_{BD} = \frac{W\sqrt{2}}{2} \cos(45^\circ) - \frac{W}{2} = -WN$$

Conditions of equilibrium at E



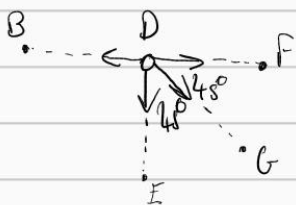
$$x: F_{EG} - F_{EC} - F_{EB} \cos(45^\circ) = 0 N$$

$$y: F_{ED} + F_{EB} \sin(45^\circ) = 0 N$$

$$F_{ED} = -\frac{W\sqrt{2}}{2} \sin(45^\circ) = -\frac{W}{2} N$$

$$F_{EG} = \frac{W\sqrt{2}}{2} \cos(45^\circ) + \frac{W}{2} = WN$$

Conditions of equilibrium at D



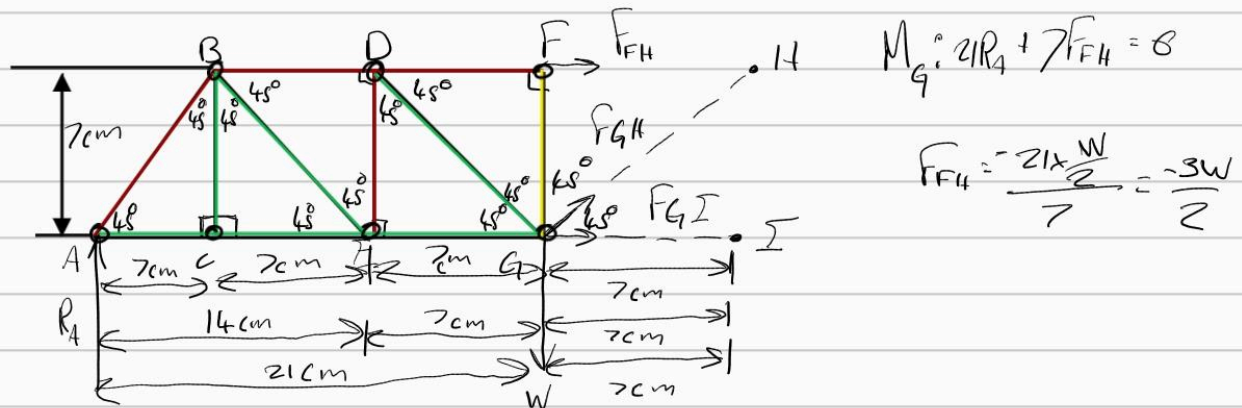
$$x: F_{DF} + F_{DG} \cos(45^\circ) = F_{DB} N$$

$$y: -F_{DE} \cos(45^\circ) - F_{DG} = 0 N$$

$$F_{DG} = \frac{-F_{DE}}{\cos(45^\circ)} = \frac{-x - \frac{W}{2}}{\cos(45^\circ)} = \frac{\sqrt{2}W}{2}$$

$$F_{FD} = -W - \frac{W\sqrt{2}}{2} \cos(45^\circ) = -\frac{3}{2}W$$

Method of sections



$$M_H: 28R_A + 7F_{GH} \sin(45^\circ) - 7F_{GH} \sin(45^\circ) + 7F_{GI} + 7W$$

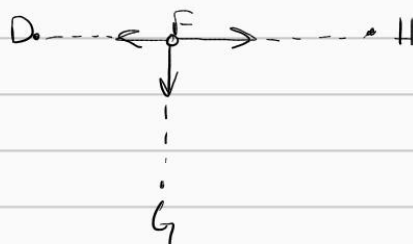
$$y: R_A + F_{GH} \sin(45^\circ) = W \quad N$$

$$F_{GH} = \frac{W - \frac{W}{2}}{\sin(45^\circ)} = \frac{W\sqrt{2}}{2} \quad N$$

$$F_{GI} = \frac{28 \times \frac{W}{2} - 7W}{7} \quad N$$

$$= W \quad N$$

Conditions of equilibrium at F



$$x: F_{FH} = F_{FD} = -\frac{3}{2}W \quad N$$

$$y: F_{FG} = 0 \quad N$$

By Symmetry from the vertical chord from where the weight acts, we can find the forces in the bars on the right hand side of the bridge (joints H, I, J, K, L).
Forces in each bar at each joint (Positive is tension, negative is ^{compression})

$$A: R_A = \frac{W}{2} N, F_{AB} = -\frac{W\sqrt{2}}{2} N, F_{AC} = \frac{W}{2} N$$

$$B: F_{BC} = 0 N, F_{BD} = -W N, F_{BF} = \frac{W\sqrt{2}}{2} N$$

$$C: F_{CE} = \frac{W}{2} N, F_{CB} = 0 N, F_{CA} = \frac{W}{2} N$$

$$D: F_{DG} = \frac{W\sqrt{2}}{2} N, F_{DF} = -\frac{3}{2} W N, F_{DE} = -\frac{W}{2} N, F_{DB} = -W N$$

$$E: F_{EG} = W N, F_{ED} = -\frac{W}{2} N, F_{EC} = \frac{W}{2} N, F_{EB} = \frac{W\sqrt{2}}{2} N$$

$$F: F_{FD} = -\frac{3}{2} W N, F_{FG} = 0 N, F_{FH} = -\frac{3}{2} W N$$

$$G: F_{GD} = \frac{W\sqrt{2}}{2} N, F_{GE} = W N, F_{GF} = 0 N, F_{GH} = \frac{W\sqrt{2}}{2} N, F_{GI} = W N$$

$$I: F_{IG} = W N, F_{IH} = -\frac{W}{2} N, F_{IK} = \frac{W}{2} N, F_{IJ} = \frac{W\sqrt{2}}{2} N$$

$$H: F_{HG} = \frac{W\sqrt{2}}{2} N, F_{HF} = -\frac{3}{2} W N, F_{HS} = -\frac{W}{2} N, F_{HJ} = -W N$$

$$K: F_{KE} = W N, F_{KJ} = 0 N, F_{KL} = \frac{W}{2} N$$

$$J: F_{JK} = 0 N, F_{JH} = -W N, F_{JI} = \frac{W\sqrt{2}}{2} N$$

$$L: R_L = \frac{W}{2} N, F_{LJ} = -\frac{W\sqrt{2}}{2} N, F_{LK} = \frac{W}{2} N$$

Force in
the most
loaded
bars of
the bridge.

Total weight of bridge = 0.08965 kg

Assume $W = 60 N \Rightarrow$ the minimum load of the bridge.

$$D: F_{DG} = 42.4 N, F_{DF} = -90 N, F_{DE} = -30 N, F_{DB} = -60 N$$

$$E: F_{EG} = 60 N, F_{ED} = -30 N, F_{EC} = 30 N, F_{EB} = 42.4 N$$

$$F: F_{FD} = -90 N, F_{FG} = 0 N, F_{FH} = -90 N$$

$$G: F_{GD} = 42.4 N, F_{GE} = 60 N, F_{GF} = 0 N, F_{GH} = 42.4 N, F_{GI} = 60 N$$

$$I: F_{IG} = 60 N, F_{IH} = -30 N, F_{IK} = 30 N, F_{IJ} = 42.4 N$$

$$H: F_{HG} = 42.4 N, F_{HF} = -90 N, F_{HS} = -30 N, F_{HJ} = -60 N$$

$6 kg \rightarrow$ min load [kg]

$181.32 \rightarrow$ predicted load/mass ratio

0.08965 kg $\rightarrow$ mass of bridge

Figure 5 and 6 (Our bridge before and after the test)





Figure 7. Results of our bridge from the test

Group No.	Box Test	Weight [g]	Weight [kg]	Load [kg]	L/W Ratio	% min Load
1	Yes	39.65	0.03965	5	126.1	71.4

6.Conclusions

The structurally efficient qualities that trusses possess prove their universal greatness in construction, especially in a classroom project like ours where students are provided a limited amount of resources to build a truss bridge that can withstand a minimum load of 60N whilst being as lightweight as possible.

It is for these reasons that we decided on constructing a Pratt truss bridge [1] “as its configuration allows for longer diagonal members and better load distribution.” which in theory, should help our bridge have the highest load/mass ratio possible and withstand the minimum load test. Its horizontal base distance is just above the 400mm minimum at 420mm and height is at 70mm which is exact to the minimum height requirement. This means that it in theory should be composed of close to the lowest mass possible that a Pratt truss bridge can have for this project which increases the load/mass ratio and overall efficiency of our bridge.

However, despite us building a bridge with the exact dimensions and layout used in our final design, as figure 7 displays its specs and load at which it failed (5kg) suggests. Perhaps if we were to do this test again, we could improve the effectiveness of our bridge by reinforcing the joints through methods such as over gluing them as when our bridge failed, it was still intact as only the lower inner joints broke it apart. We could’ve also reinforced our joints by cutting parts of our energy board to make lightweight gusset plates that attach at each joint which would ensure proper load transfer at joints whilst simultaneously securely fastening the joints together. This means our bridge would be less prone to failure and able to carry more load whilst also having a lower mass than if we just decided to implement more trusses.

6. References

[1] Watane, S. and Dahat, I. (2023). *REVIEW STUDY ON PRATT AND HOWE TRUSS*. [online] *International Research Journal of Modernization in Engineering Technology and Science* (*Peer-Reviewed*, p.7. Available at: https://www.irjmets.com/uploadedfiles/paper//issue_7_july_2023/43146/final/fin_irjmets1689347630.pdf.